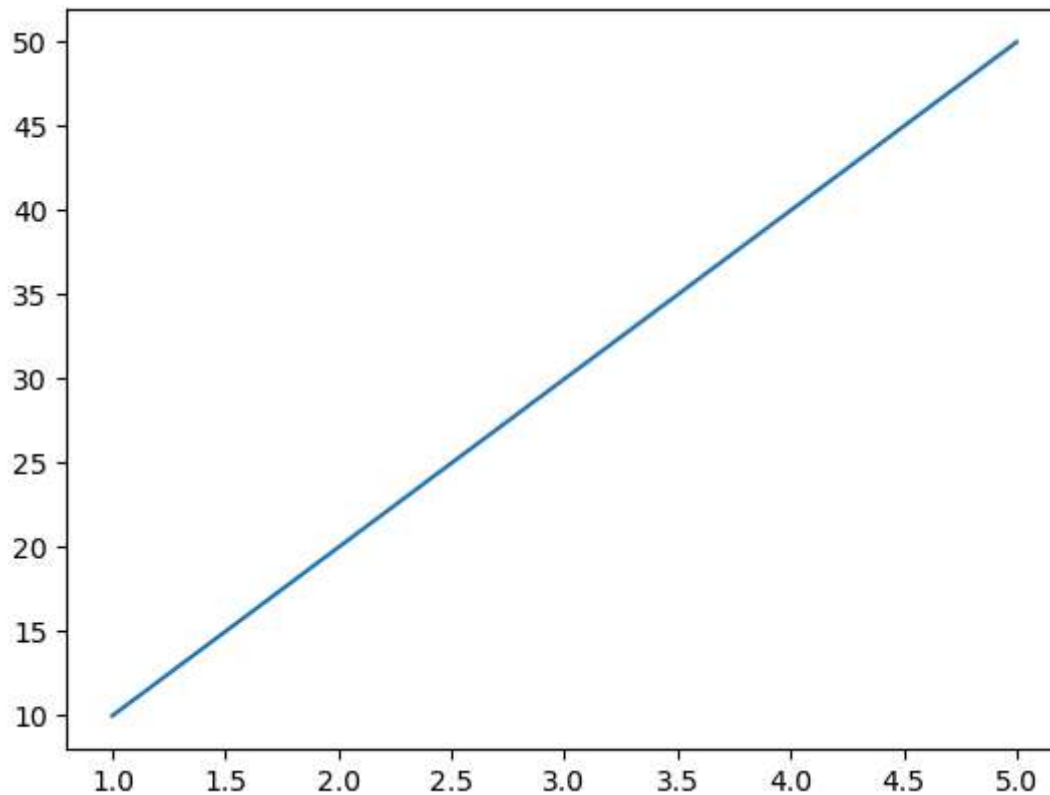


```
In [2]: !pip install matplotlib
```

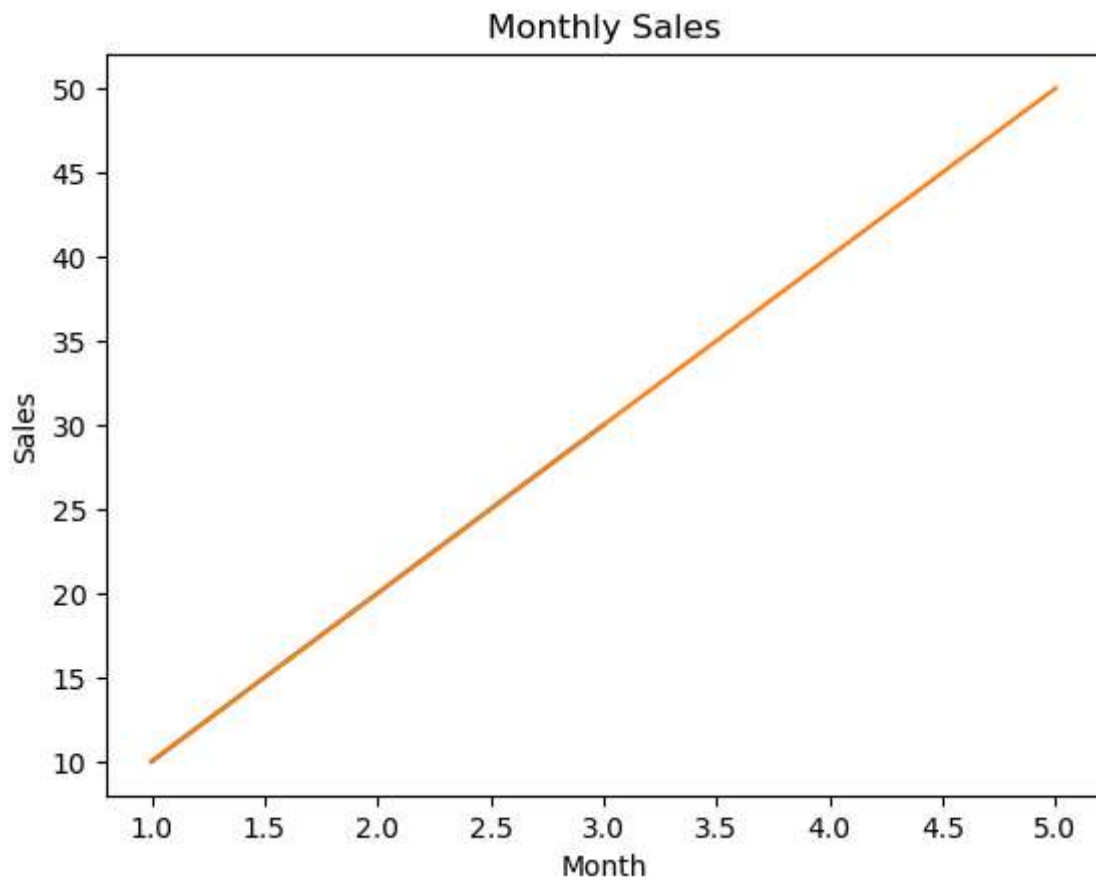
```
Requirement already satisfied: matplotlib in c:\users\arthi\anaconda3\lib\site-packa
ges (3.10.6)
Requirement already satisfied: contourpy>=1.0.1 in c:\users\arthi\anaconda3\lib\site
-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cyclor>=0.10 in c:\users\arthi\anaconda3\lib\site-pac
kages (from matplotlib) (0.11.0)
Requirement already satisfied: fonttools>=4.22.0 in c:\users\arthi\anaconda3\lib\sit
e-packages (from matplotlib) (4.60.1)
Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\arthi\anaconda3\lib\sit
e-packages (from matplotlib) (1.4.9)
Requirement already satisfied: numpy>=1.23 in c:\users\arthi\anaconda3\lib\site-pack
ages (from matplotlib) (2.3.5)
Requirement already satisfied: packaging>=20.0 in c:\users\arthi\anaconda3\lib\site-
packages (from matplotlib) (25.0)
Requirement already satisfied: pillow>=8 in c:\users\arthi\anaconda3\lib\site-packag
es (from matplotlib) (12.0.0)
Requirement already satisfied: pyparsing>=2.3.1 in c:\users\arthi\anaconda3\lib\site
-packages (from matplotlib) (3.2.5)
Requirement already satisfied: python-dateutil>=2.7 in c:\users\arthi\anaconda3\lib
\site-packages (from matplotlib) (2.9.0.post0)
Requirement already satisfied: six>=1.5 in c:\users\arthi\anaconda3\lib\site-package
s (from python-dateutil>=2.7->matplotlib) (1.17.0)
```

```
In [3]: import matplotlib.pyplot as plt
x=[1,2,3,4,5]
y=[10,20,30,40,50]
plt.plot(x,y)
plt.show()
```



```
In [4]: #figure and Axes  
#There are must important concepts.  
#Figure - The Entire Chart window.  
#Axes - The Area where the group is drawn
```

```
In [5]: import matplotlib.pyplot as plt  
fig,ax = plt.subplots()  
ax.plot([1,2,3],[10,20,30])  
#Adding a Title  
plt.title('Monthly Sales')  
#Adding X and Y Lables  
plt.xlabel("Month")  
plt.ylabel("Sales")  
plt.plot(x,y)  
plt.show()
```

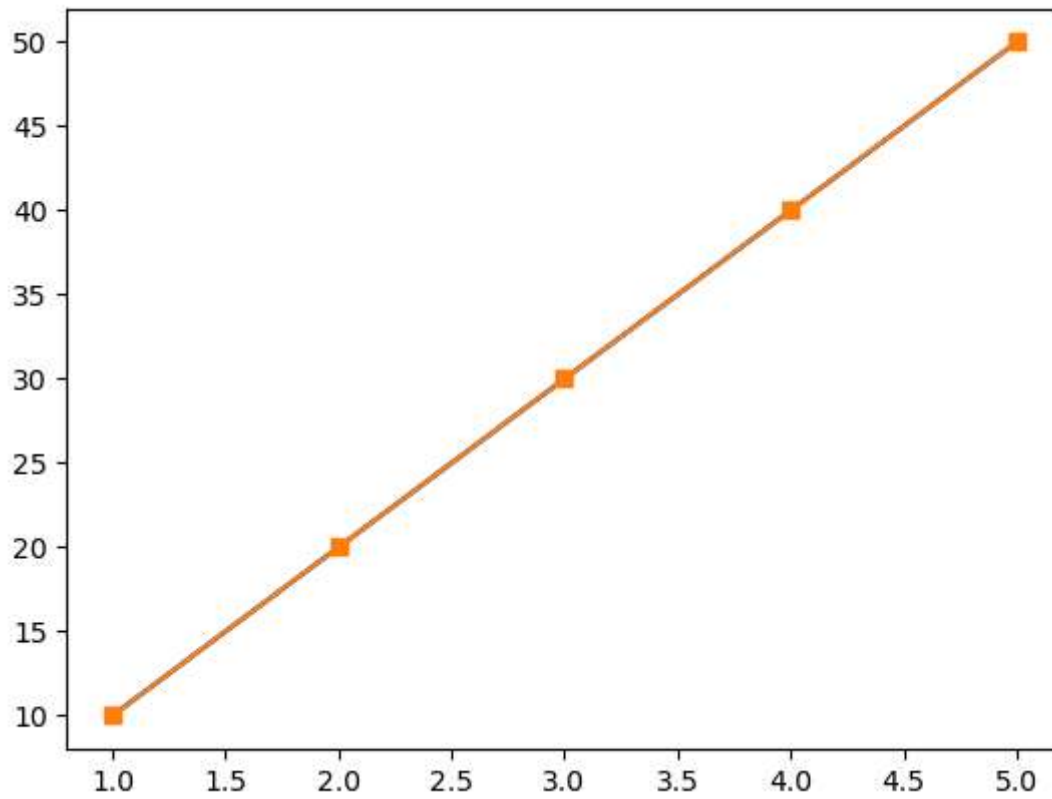


```
In [6]: #Changing Line Color
plt.plot(x,y,color='blue')

#changing line style
plt.plot(x,y,linestyle='--')

#Adding Markers
plt.plot(x,y,marker='s')
```

```
Out[6]: [<matplotlib.lines.Line2D at 0x243a2877610>]
```



```
In [7]: #o -circle  
#s -Square  
#^ -triangle  
#* - Star
```

```
In [8]: !pip install seaborn
```

Requirement already satisfied: seaborn in c:\users\arthi\anaconda3\lib\site-packages (0.13.2)

Requirement already satisfied: numpy!=1.24.0,>=1.20 in c:\users\arthi\anaconda3\lib\site-packages (from seaborn) (2.3.5)

Requirement already satisfied: pandas>=1.2 in c:\users\arthi\anaconda3\lib\site-packages (from seaborn) (2.3.3)

Requirement already satisfied: matplotlib!=3.6.1,>=3.4 in c:\users\arthi\anaconda3\lib\site-packages (from seaborn) (3.10.6)

Requirement already satisfied: contourpy>=1.0.1 in c:\users\arthi\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (1.3.3)

Requirement already satisfied: cycler>=0.10 in c:\users\arthi\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (0.11.0)

Requirement already satisfied: fonttools>=4.22.0 in c:\users\arthi\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (4.60.1)

Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\arthi\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (1.4.9)

Requirement already satisfied: packaging>=20.0 in c:\users\arthi\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (25.0)

Requirement already satisfied: pillow>=8 in c:\users\arthi\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (12.0.0)

Requirement already satisfied: pyparsing>=2.3.1 in c:\users\arthi\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (3.2.5)

Requirement already satisfied: python-dateutil>=2.7 in c:\users\arthi\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (2.9.0.post0)

Requirement already satisfied: pytz>=2020.1 in c:\users\arthi\anaconda3\lib\site-packages (from pandas>=1.2->seaborn) (2025.2)

Requirement already satisfied: tzdata>=2022.7 in c:\users\arthi\anaconda3\lib\site-packages (from pandas>=1.2->seaborn) (2025.2)

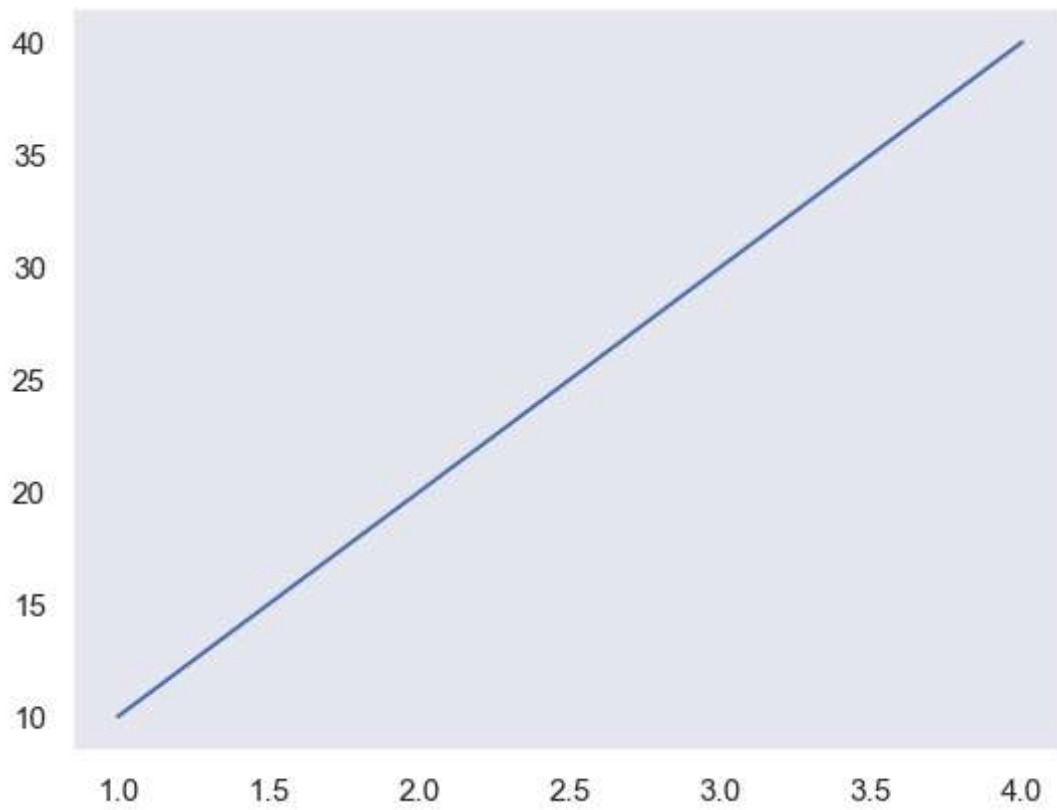
Requirement already satisfied: six>=1.5 in c:\users\arthi\anaconda3\lib\site-packages (from python-dateutil>=2.7->matplotlib!=3.6.1,>=3.4->seaborn) (1.17.0)

```
In [9]: import seaborn as sns
import matplotlib.pyplot as plt
x = [1,2,3,4]
y = [10,20,30,40]

#Themes in Seaborn
#Themes control the appearance of charts
#Default Themes
sns.set_theme()

#Dark Grid Theme
sns.set_style('dark')

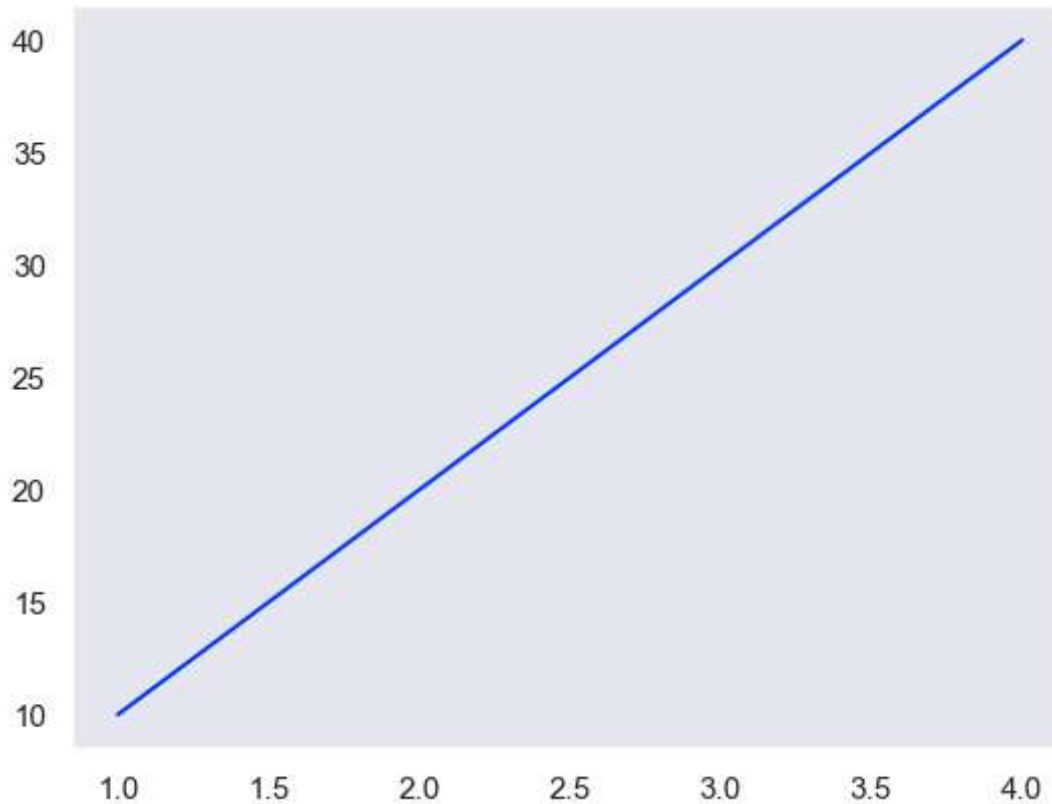
#Available Styles
#DarkGrid
#White grid
#Dark
#white
sns.lineplot(x=x,y=y)
plt.show()
```



```
In [10]: #A palettes is a collection of colors  
#View Color Palatte
```

```
#Available Colors  
#deep  
#Dark  
#Bright  
#Muted  
#palette  
#colorblind
```

```
sns.set_palette("bright")  
sns.lineplot(  
    x=[1,2,3,4],  
    y=[10,20,30,40]  
)  
plt.show()
```



```
In [11]: #Build in dataset support
#One of Seaborn's biggest advantages is build in datasets.
#view Available Dataset
import seaborn as sns
print(sns.get_dataset_names())

['anagrams', 'anscombe', 'attention', 'brain_networks', 'car_crashes', 'diamonds',
'dots', 'dowjones', 'exercise', 'flights', 'fmri', 'geyser', 'glue', 'healthexp', 'i
ris', 'mpg', 'penguins', 'planets', 'seance', 'taxis', 'tips', 'titanic']
```

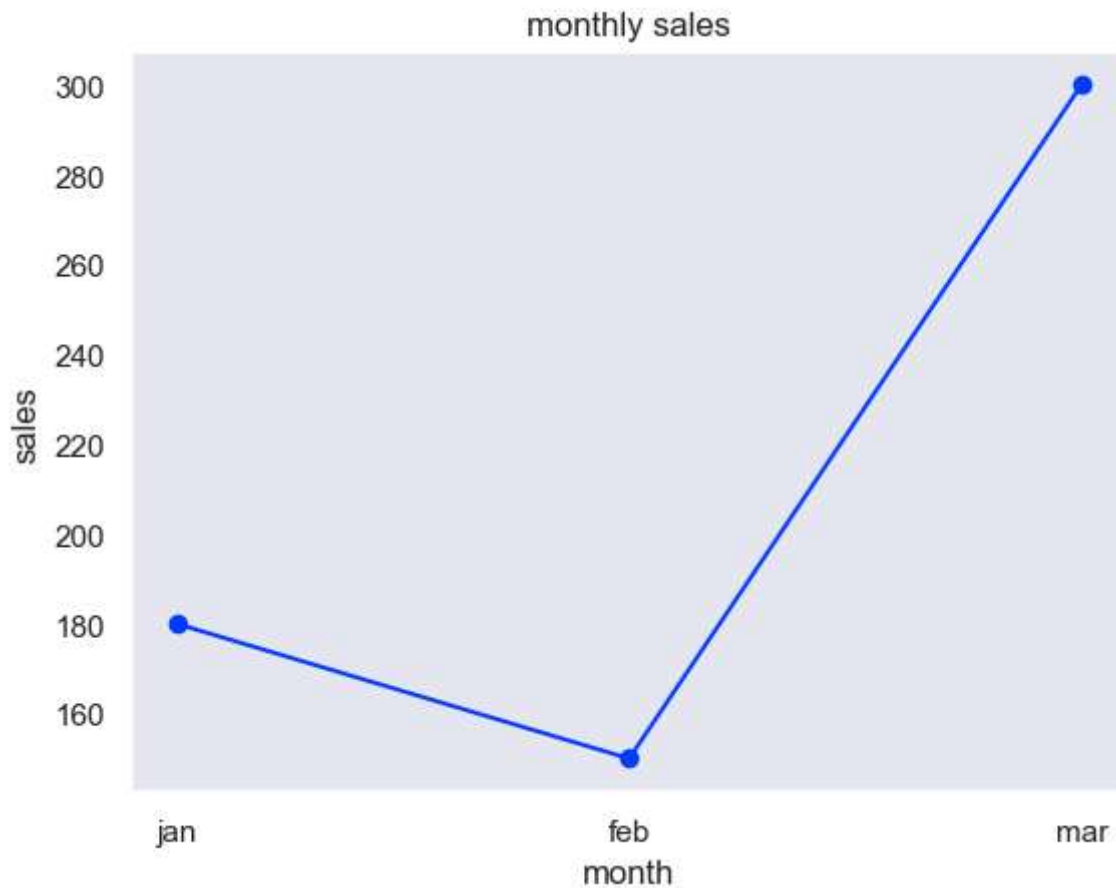
```
In [12]: #Load titanic dataset
import seaborn as sns
df = sns.load_dataset('titanic')
print(df.head())
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	\
0	0	3	male	22.0	1	0	7.2500	S	Third	
1	1	1	female	38.0	1	0	71.2833	C	First	
2	1	3	female	26.0	0	0	7.9250	S	Third	
3	1	1	female	35.0	1	0	53.1000	S	First	
4	0	3	male	35.0	0	0	8.0500	S	Third	

	who	adult_male	deck	embark_town	alive	alone
0	man	True	NaN	Southampton	no	False
1	woman	False	C	Cherbourg	yes	False
2	woman	False	NaN	Southampton	yes	True
3	woman	False	C	Southampton	yes	False
4	man	True	NaN	Southampton	no	True

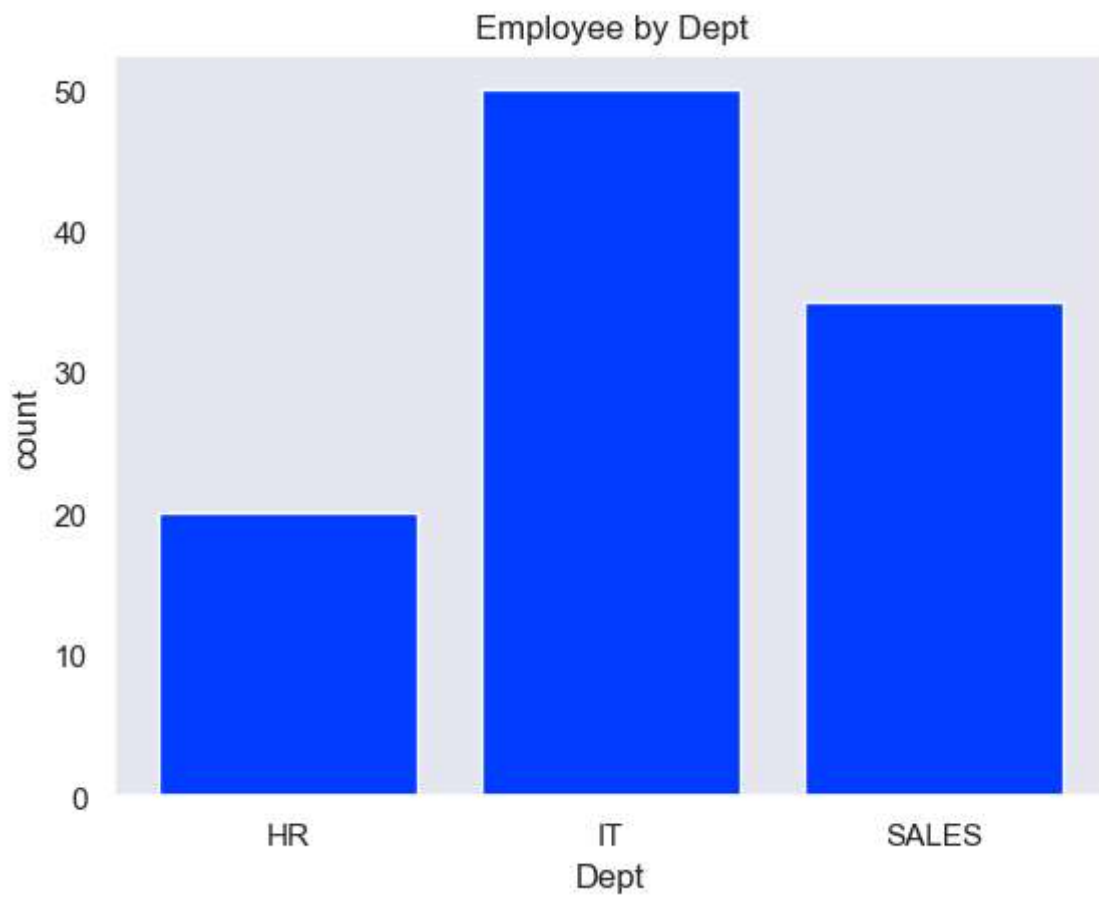
```
In [13]: #Line charts
#A line charts connect data points with lines
```

```
import matplotlib.pyplot as plt
month = ['jan', 'feb', 'mar']
sales=[180,150,300]
plt.plot(month,sales,marker='o')
plt.title("monthly sales")
plt.xlabel("month")
plt.ylabel("sales")
plt.show()
```

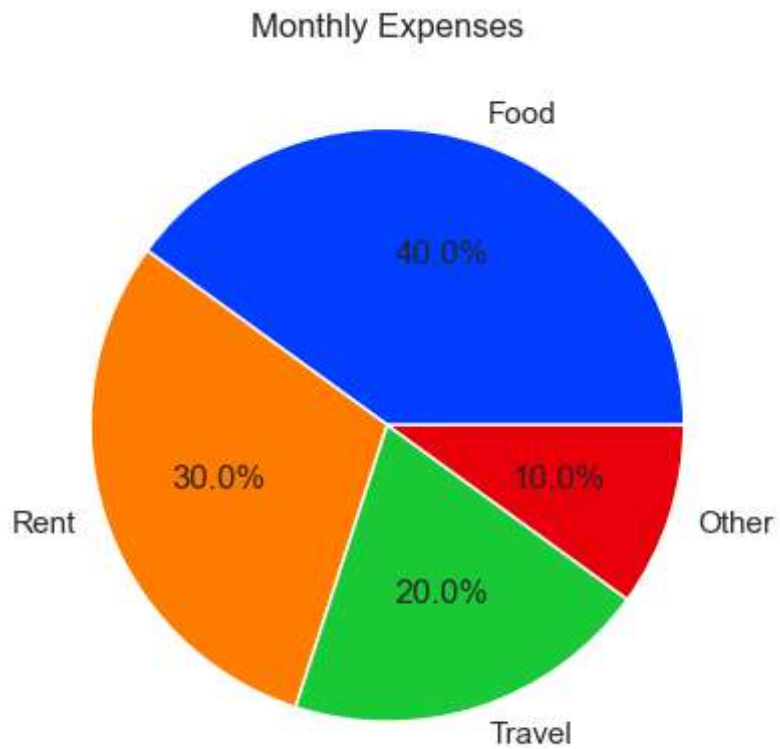


```
In [14]: #Compare Values
import matplotlib.pyplot as plt
dept=['HR','IT','SALES']
employees = [20,50,35]

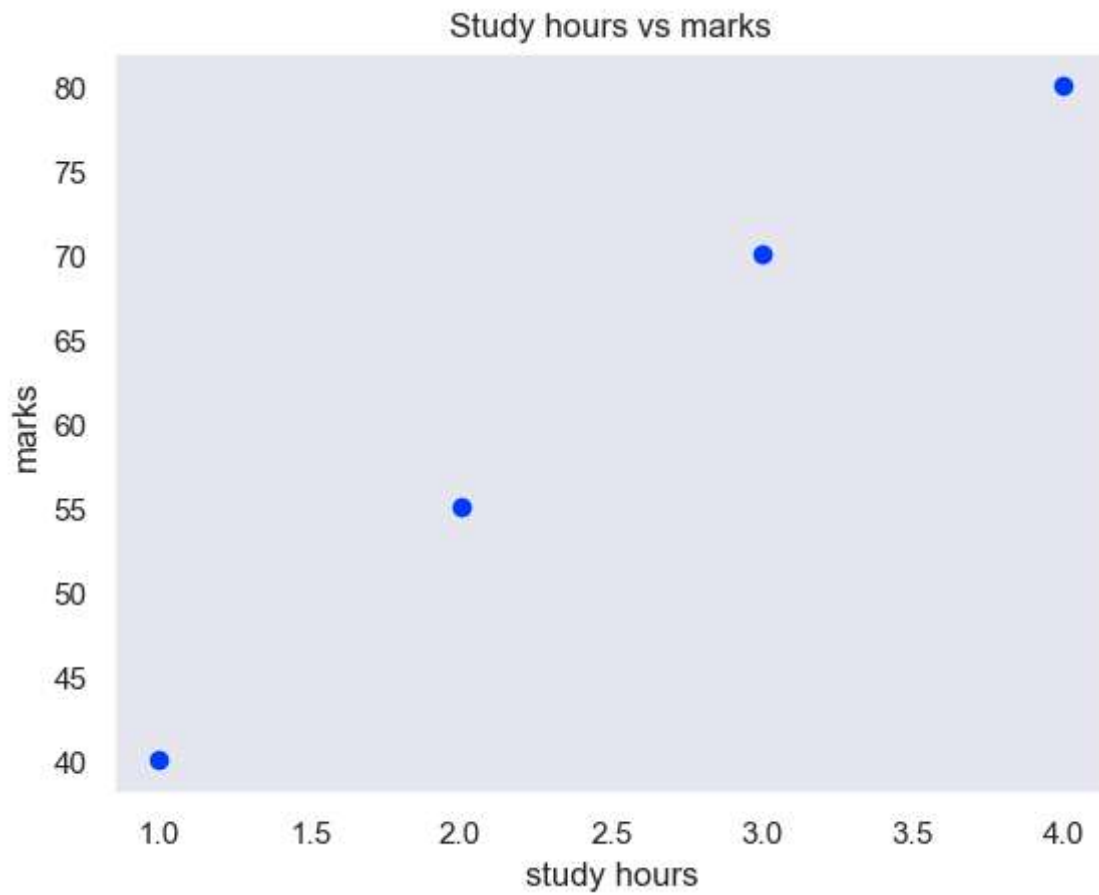
plt.bar(dept,employees)
plt.title("Employee by Dept")
plt.xlabel("Dept")
plt.ylabel("count")
plt.show()
```

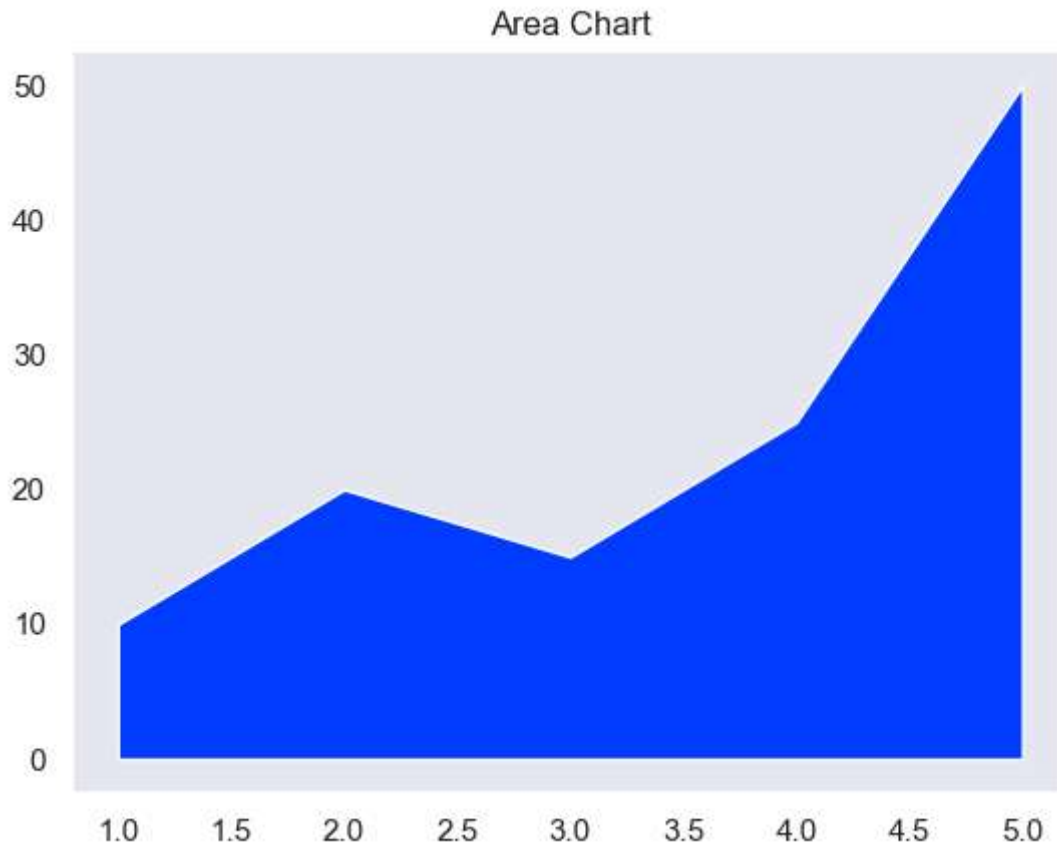
```
In [15]: import matplotlib.pyplot as plt
labels = ['Food', 'Rent', 'Travel', 'Other']
values = [40, 30, 20, 10]
plt.pie(values, labels = labels, autopct = "%1.1f%%")
plt.title("Monthly Expenses")
plt.show()
```



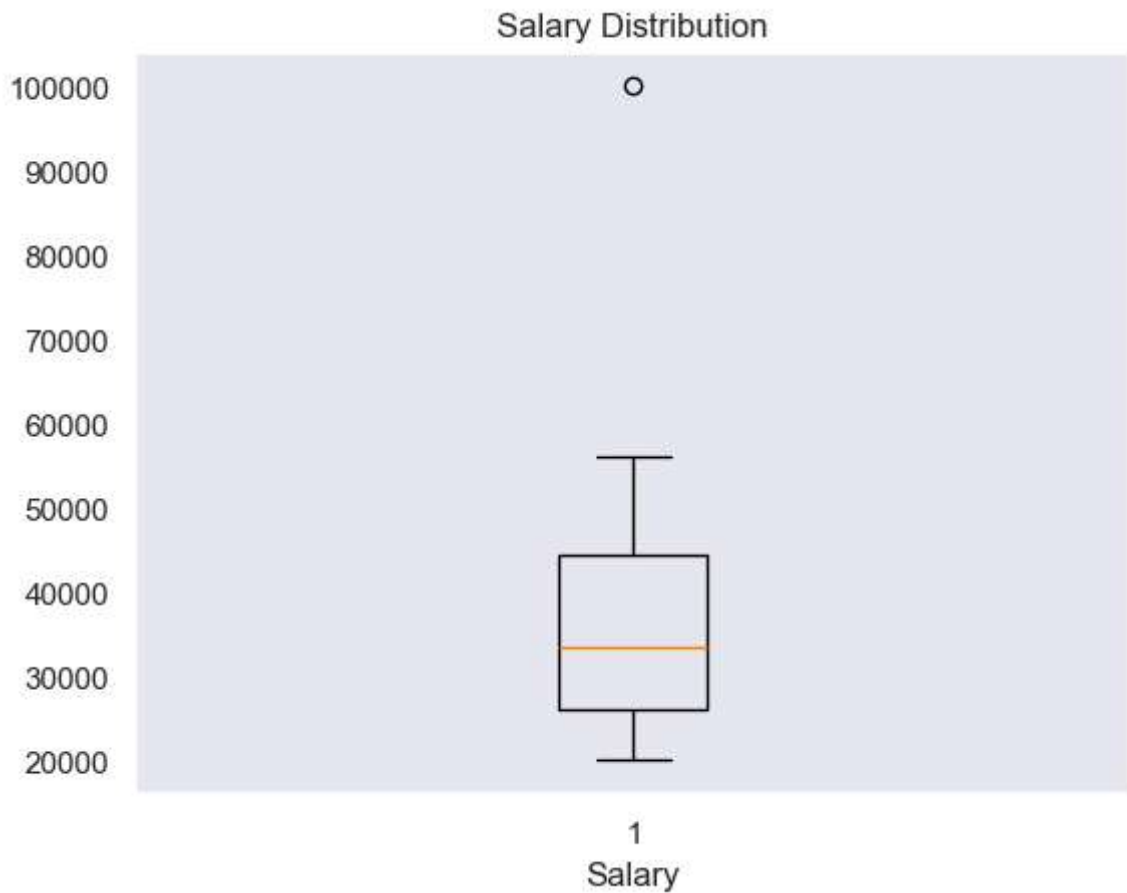
```
In [17]: #Scatter plot
import matplotlib.pyplot as plt
hours=[1,2,3,4]
marks=[40,55,70,80]
plt.scatter(hours,marks)
plt.title("Study hours vs marks")
plt.xlabel("study hours")
plt.ylabel("marks")
plt.show()
```



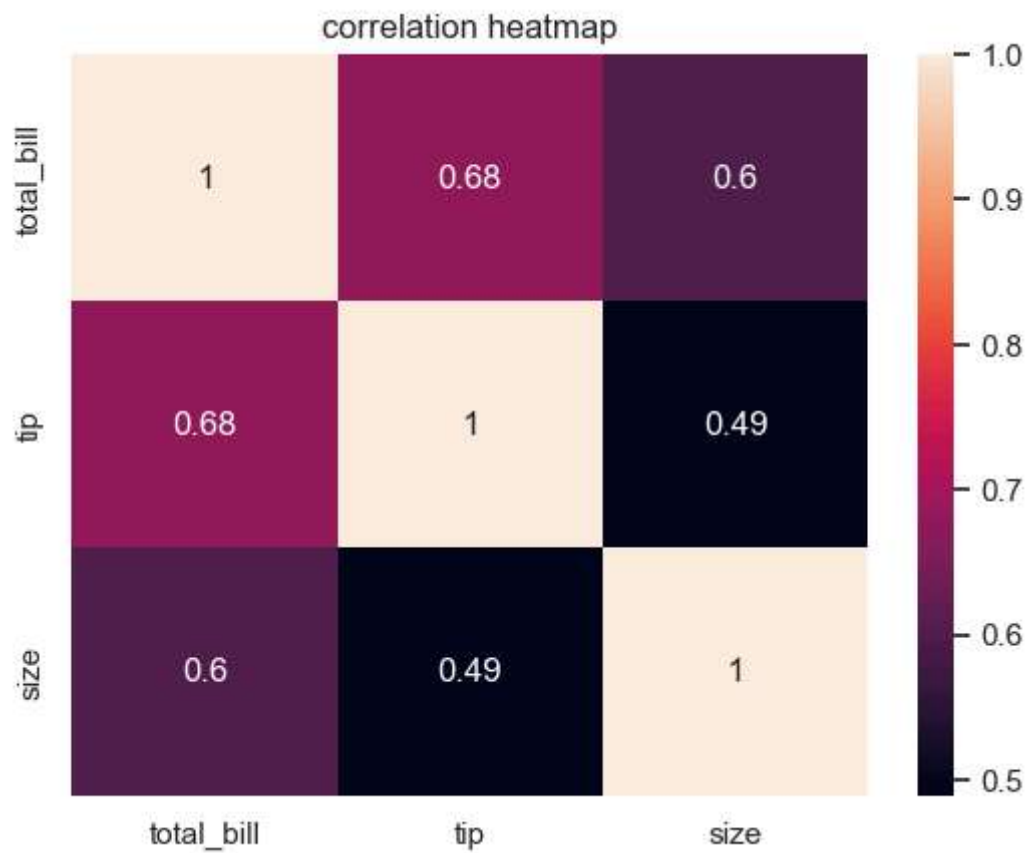
```
In [19]: #Area chart
import matplotlib.pyplot as plt
x = [1,2,3,4,5]
y=[10,20,15,25,50]
plt.fill_between(x,y)
plt.title("Area Chart")
plt.show()
```



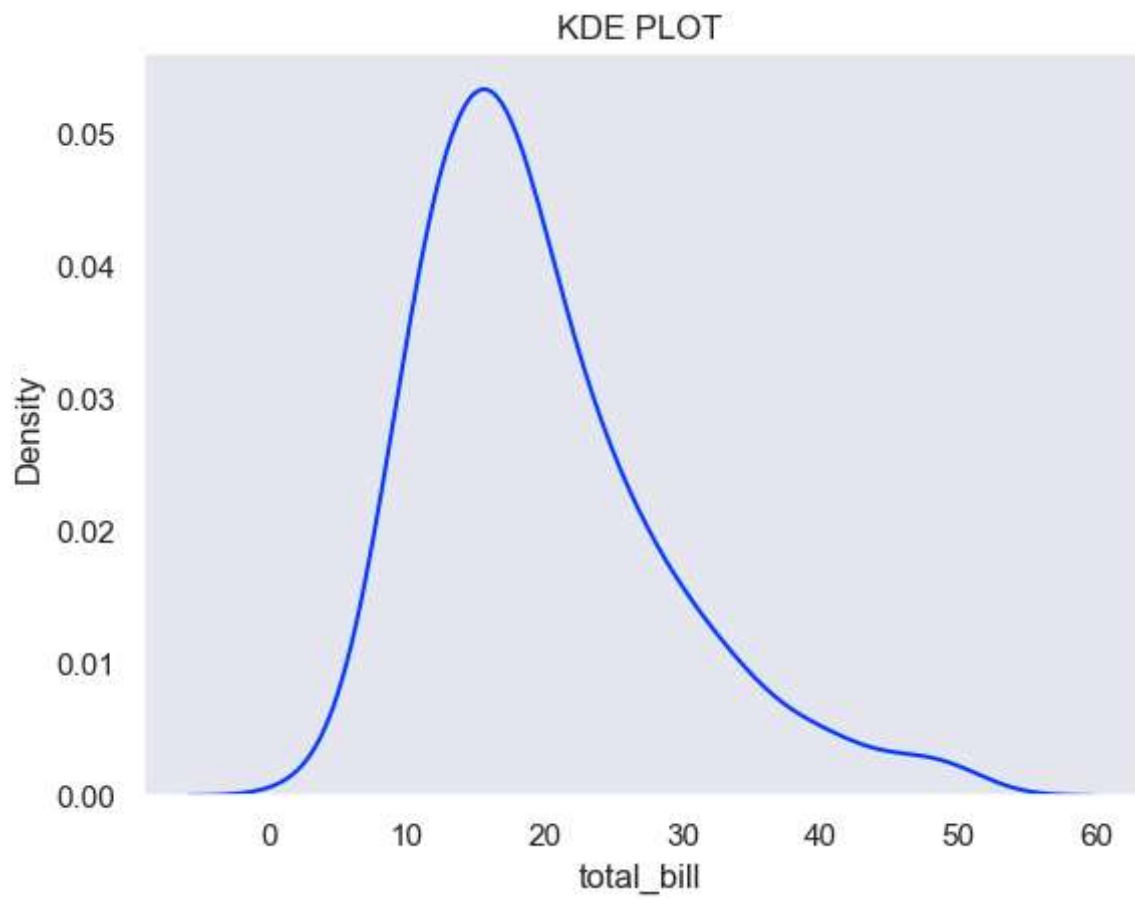
```
In [20]: #Box plot is used to visualize the distribution of numrical data and identify outl  
import matplotlib.pyplot as plt  
salary =[20000,25000,30000,43000,56000,  
          37000,26000,45000,26000,100000]  
plt.boxplot(salary)  
plt.title("Salary Distribution")  
plt.xlabel("Salary")  
plt.show()
```



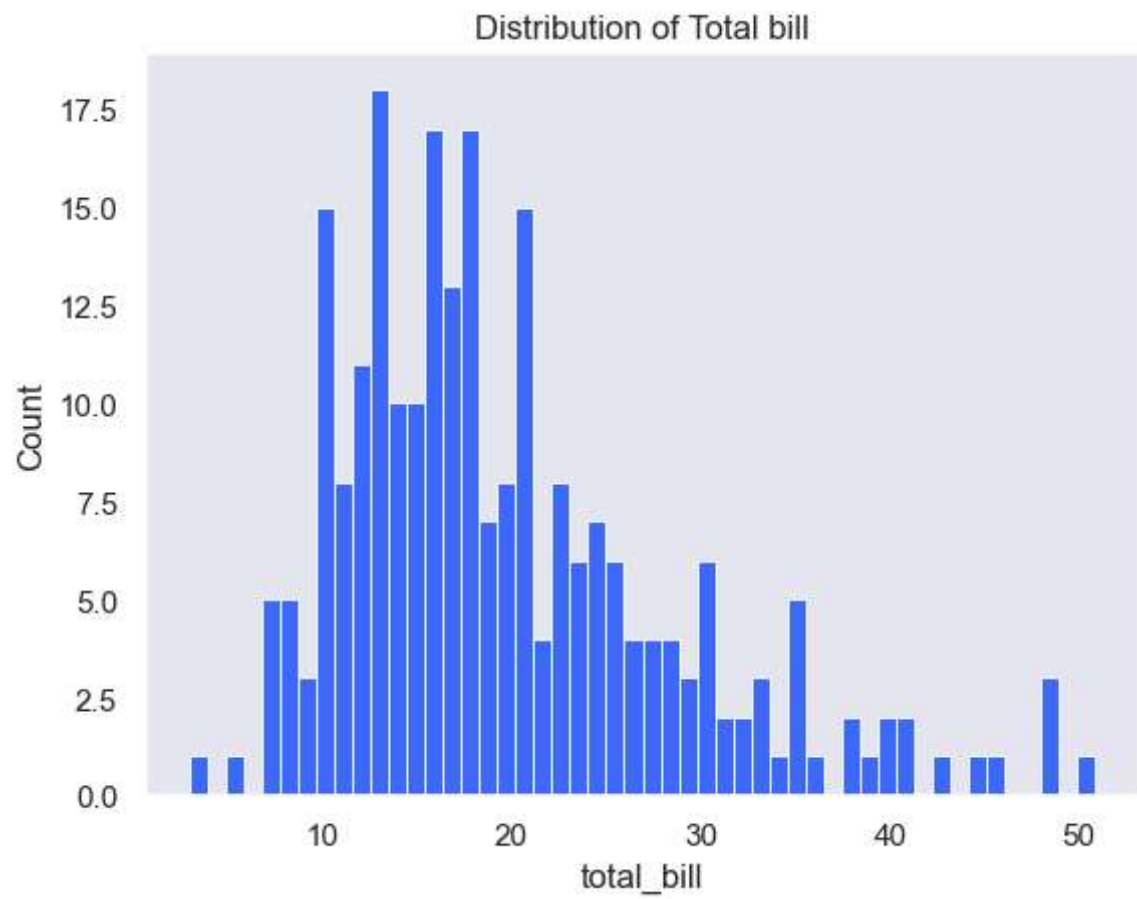
```
In [21]: #Heat map uses colors to represent values in a table o matrix.  
#In data analysis heatmap are commonly used to visualize correlation between numeri  
  
import matplotlib.pyplot as plt  
import seaborn as sns  
  
#Load Dataset  
tips = sns.load_dataset("tips")  
  
#correlation matrix  
corr = tips.corr(numeric_only=True)  
  
#heatmap  
sns.heatmap(corr,annot=True)  
  
plt.title("correlation heatmap")  
plt.show()
```



```
In [22]: #Kernel Density Estimation
#smooth version of a histogram
#instead of bars,it shows a smooth curve representing data distribution.
sns.kdeplot(data=tips,x="total_bill")
plt.title("KDE PLOT")
plt.show()
```



```
In [23]: #Histogram using seaborn
sns.histplot(data=tips,x="total_bill",bins=50)
plt.title("Distribution of Total bill")
plt.show()
```



```
In [24]: #sns.histogram()  
#Create Histogram.
```

```
In [ ]:
```

```
In [ ]:
```